



# Modification and mediation effects of ozone on heatwave-mortality association: A time series study in five provinces of China<sup>☆</sup>

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## ABSTRACT

Epidemiological studies have demonstrated that heatwaves are associated with increased mortality risk. However, knowledge about modifying and mediating effects of O<sub>3</sub> on this relationship is still insufficient. Daily mortality data for the summer of 2013–2017 were obtained from 278 locations across China. Daily meteorological data for the same period were sourced from the fifth generation of the European reanalysis (ERA5), while daily O<sub>3</sub> data were obtained from the China High Air Pollutants (CHAP) dataset. The modification effects of O<sub>3</sub> on the heatwave-mortality associations were estimated using multiplicative interaction and additive interaction analyses. A mediation model was employed to assess the mediating effect of O<sub>3</sub> on the association. The excess risk (ER) of heatwave on mortality was 9.17 % (95 % confidence interval [CI]: 7.91–10.45 %). Heatwaves were associated with a higher mortality risk during periods of high O<sub>3</sub> pollution (ER = 9.68 %, 95 % CI: 8.35–11.05 %) than periods of low O<sub>3</sub> pollution (ER = 5.96 %, 95 % CI: 4.23–7.74 %). Additionally, we observed a 2.70 % (95 % CI: 0.20–5.20 %) additive mortality interaction between O<sub>3</sub> and heatwaves for the total population, with much higher values among individuals aged ≥85 years (6.80 %, 95 % CI: 2.50–10.90 %) and those with cardiovascular diseases (4.20 %, 95 % CI: 0.60–7.70 %). Moreover, we found that 9.33 % (95 % CI: 7.17–11.76 %) of the heatwave effect on mortality was mediated by O<sub>3</sub>, with a higher value observed in individuals aged ≥85 years (11.48 %, 95 % CI: 8.94–14.33 %). O<sub>3</sub> modified and mediated the heatwave-mortality association. Our results provide evidence that O<sub>3</sub> may function in a dual capacity: partly mediating the effect of heatwave on mortality and partly modifying the strength of this association. This dual role reflects the complex interplay between meteorological conditions, air quality, and health outcomes.

## 1. Introduction

With climate change, heatwaves have become increasingly frequent extreme weather events (Guo et al., 2018; Pu et al., 2017) and are recognised as a major global health threat in the 21st century (Ma et al.,

2015; Pan et al., 2023; Yan et al., 2022). Unlike single-day high-temperature events, a heatwave consists of multiple consecutive days of elevated temperatures, which may pose a more significant threat to public health than a brief heat episode. For instance, a study based on 130 Chinese counties found that the heatwaves lasting two or three

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consecutive days carry a greater risk of death than a one-day extreme-temperature event (Sun et al., 2021). The heatwave has placed a significant health burden on the population. A global study that included 43 countries/territories found that more than 153,000 deaths attributed to heatwaves per warm season from 1990 to 2019, accounting for 0.94 % of all deaths per warm season (Zhao et al., 2024). In Europe, a heatwave in 2022 is reported to have caused 61,672 heat-related deaths. In 2022, China experienced the most severe heatwave since 1961, which led to a record-breaking heatwave-related mortality of approximately 50,900 deaths, more than twice the number in 2021 (Zhang et al., 2023).

Increased heatwaves are strongly associated with ozone (O<sub>3</sub>) concentrations, as they share common underlying meteorological drivers and usually coincide (Ortu et al., 2013). For instance, elevated O<sub>3</sub> concentrations were observed in the UK and France during a heatwave in August 2003 (Dear et al., 2005; Johnson et al., 2005). In China, previous study observed that the probability of maximum daily 8-h average (MDA8) O<sub>3</sub> exceedance increased from 62.40 % on non-heatwave days to 82.50 % on heatwave days (Li et al., 2024). Another study in Sichuan of China also found that every 1 °C increase in air temperature was related to 4.12–5.40 ppb increase of O<sub>3</sub> (Wang et al., 2024). O<sub>3</sub> is a major atmospheric pollutant that poses substantial risks to human health (Guo et al., 2023; Zhang et al., 2021b), and a Global Burden of Disease study estimated 400 thousand deaths attributed to O<sub>3</sub> pollution in 2019 (Murray et al., 2020).

Given the interaction between heatwave and O<sub>3</sub>, as compound heatwave-O<sub>3</sub> pollution events are expected to become more frequent and intense owing to climate change. Therefore, people may face greater health risks from simultaneous exposure to both high air temperatures and elevated O<sub>3</sub> concentrations, as these two stressors may interact to impact human health. O<sub>3</sub> and heatwaves reportedly share common physiological mechanisms and have combined effects. Heatwaves and O<sub>3</sub> can synergistically increase the risk of cardiovascular events by exacerbating existing conditions and promoting inflammation (Cheng et al., 2024; Goodman et al., 2015). They can also indirectly promote oxidative stress, which can damage cells and tissues, including those in the respiratory and cardiovascular systems (Cheng et al., 2024; Wiegman et al., 2020). Several previous studies have investigated the synergistic effects of heatwave exposure and high O<sub>3</sub> (Analitis et al., 2018; Analitis et al., 2014). For instance, a recent study reported significant synergistic effects on cardiovascular disease (CVD) mortality following exposure to both high O<sub>3</sub> and heatwaves (Xu et al., 2023). A study in Chengdu (China) has also indicated that high temperature could further aggravate the mortality risks of O<sub>3</sub> (Zhang et al., 2022). However, this study only focused on CVD in a single region in China; therefore, its findings have limited generalizability. Therefore, a nationwide study is required to comprehensively investigate interaction effect between heatwave and O<sub>3</sub> on health.

Previous studies on the heatwave-mortality association have conceptualised the role of O<sub>3</sub> as either a confounder or an effect modifier (Reid et al., 2012; Xu et al., 2023). Nevertheless, as air temperature is positively associated with the rate of photochemical processes in O<sub>3</sub> formation, O<sub>3</sub> may act as a potential mediator of the acute health effects of heatwaves on mortality. Despite this, only a few studies have investigated the mediating role of O<sub>3</sub> on the heatwave-mortality association, such as research conducted in South Korea and France (Alari et al., 2023; Bae et al., 2023). Additionally, in China, people are being confronted with an increase in both heatwave and high O<sub>3</sub> pollution. Therefore, clarifying the role of O<sub>3</sub> could help identify key paths in the causal relationship between heatwave and mortality and also provide evidence on how to enhance adaptation to concurrent heatwaves and O<sub>3</sub> pollution.

As reported in previous studies, O<sub>3</sub> pollution in China exhibits pronounced regional disparities, with Central and Eastern China as well as the Southeast Coastal Area experiencing both higher concentrations and more rapid increases in recent years (Meng et al., 2022). The spatial heterogeneous of heatwave was also observed in China that the largest

heatwave frequency happened to be mainly in Southern China, and Northern China (Liu et al., 2021a). Herein, we conducted a multi-region study in southern, central, eastern, southwestern, and northern provinces (278 counties/districts) of China (2013–2017) to examine the mortality effect of heatwave exposure and explore the modifying and mediating effects of O<sub>3</sub> on the heatwave-mortality association. Our findings may help elucidate the mechanism by which compound heatwave-O<sub>3</sub> events affect mortality, providing valuable insights for coordinating climate change management and air pollution control.

## 2. Methods

### 2.1. Study location selection

Based on the regional distribution in China, mortality surveillance points (districts/counties) in five provinces, including Guangdong Province in Southern China, Hunan Province in Central China, Zhejiang Province in Eastern China, Yunnan Province in Southwestern China, and Jilin Province in Northern China, were selected for this study (Fig. 1). To ensure that adequate statistical power was achieved for the death count in the model for estimating effects, according to previous studies (Hu et al., 2021; Liu et al., 2021b), counties/districts were screened based on population size and mortality rate. Districts/counties that met either of the following criteria were included: (1) a population count greater than 200,000; (2) an annual mortality rate greater than 4 % (Liu et al., 2021b; Ma et al., 2015). These five provinces covered more than 321 million people in 2017, accounting for 23.0 % of China's population (National Bureau of Statistics of China, 2022).

### 2.2. Data collection

#### 2.2.1. Mortality data

The daily number of deaths in summer (1 June–31 August) during 2013–2017 in 278 counties/districts were collected from the mortality surveillance system at the corresponding Provincial Centre of Disease Control and Prevention. The standards for data collection by the mortality surveillance system are unified nationwide and have been described in a previous literature (Liu et al., 2016). Causes of death were coded according to the International Classification of Diseases tenth Revision (ICD-10), including non-accidental causes (A00–R99), CVDs (I00–I99), cerebrovascular diseases (CBD, I60–I69), and respiratory diseases (J00–J99). The daily number of non-accidental deaths was further stratified by sex (male and female), age (0–64 years, 65–84 years, and ≥85 years) and region (Southern, Central and Northern).

#### 2.2.2. Environmental data

Daily maximum air temperature (°C), daily mean air temperature (°C), and relative humidity (%) data during 2013–2017 were derived from the fifth generation of European reanalysis (ERA5), which is a global climate-atmosphere reanalysis dataset, provided by the European Centre for Medium-Range Weather Forecasts, covering the period from January 1950 to the present (Royé et al., 2020). This dataset provides a conventional latitude and longitude grid with a resolution of approximately 25 km × 25 km (0.25° × 0.25°), and has been validated in a previous study (Yilmaz, 2023). More detailed information is shown in Fig. S2.

Daily concentrations of O<sub>3</sub> (8-h maximum O<sub>3</sub>), PM<sub>2.5</sub> (particulate matter with an aerodynamic diameter ≤2.5 μm), CO (carbon monoxide), NO<sub>2</sub> (nitrogen dioxide), SO<sub>2</sub> (sulphur dioxide), and PM<sub>10</sub> (particulate matter with an aerodynamic diameter ≤10.0 μm) were obtained from the China High Air Pollutants (CHAP) datasets (<http://www.geodata.cn>). The CHAP is a set of raster data with a resolution of 1 km × 1 km, and ten-fold cross-validation indicates good accuracy (Wei et al., 2022; Wei et al., 2021a; Wei et al., 2023; Wei et al., 2021b).

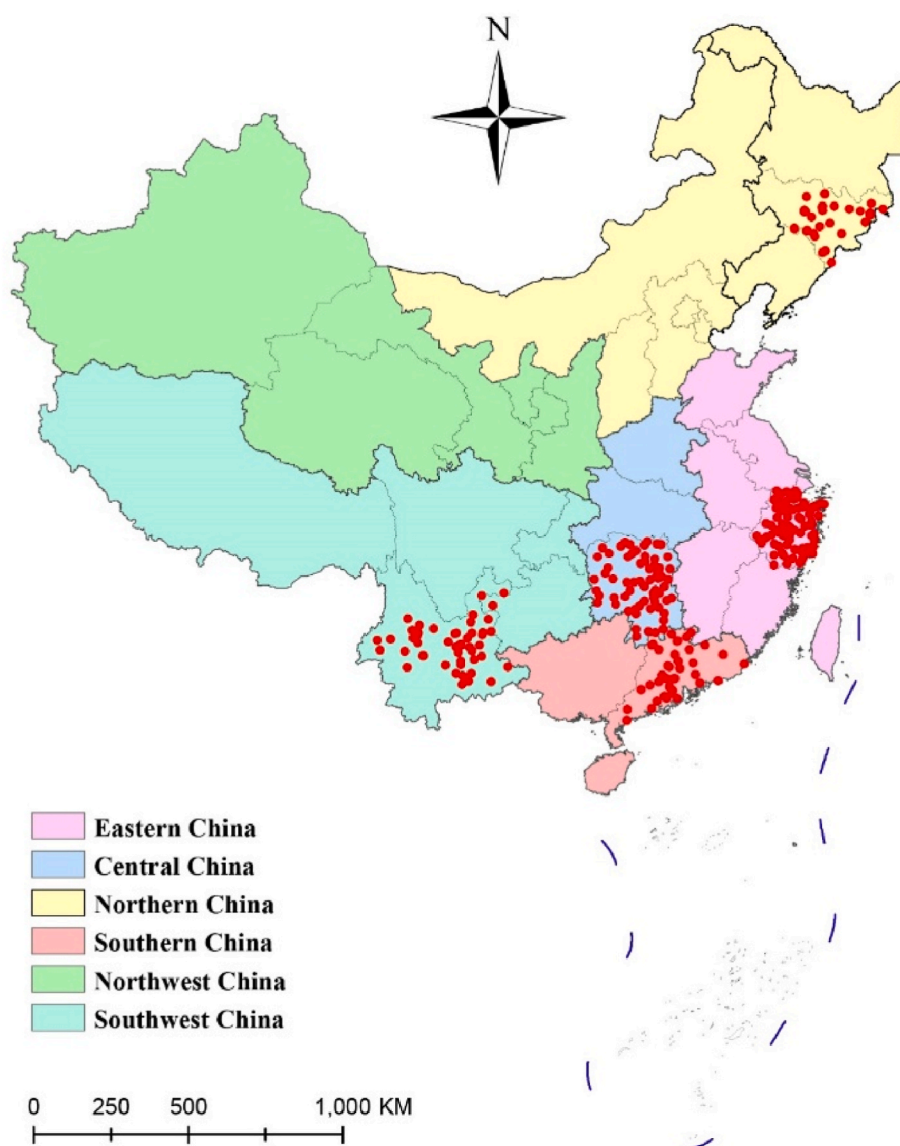


Fig. 1. Geographic distribution of 278 study counties/districts in China.

## 2.3. Statistical analysis

### 2.3.1. Defining heatwave

According to previous studies (Chen et al., 2022; Yang et al., 2019), we defined a heatwave as a period with a daily maximum air temperature  $\geq 92.5^{\text{th}}$  percentile of daily maximum temperatures during 2013–2017, lasting at least three consecutive days. To examine the health effects of heatwave intensity and duration, we also used the 90.0<sup>th</sup>, 92.5<sup>th</sup>, and 95.0<sup>th</sup> percentiles of daily maximum air temperature as thresholds and defined heatwaves based on durations of 2, 3, and 4 consecutive days (Table S1). Because heatwaves mainly occur during the warm season, we limited the study period to June–August.

### 2.3.2. Examining the independent association of heatwave with mortality

We employed a widely used and validated two-stage approach to estimate the independent association between heatwave, and mortality (Fang et al., 2023; Liu et al., 2018). In the first stage, we used a generalised linear model with a quasi-Poisson link function to control overdispersion and analyse the county-/district-specific exposure-response relationship between heatwaves and mortality. The equation used is as follows:

$$\text{Log}[E(Y_t)] = a + HW + ns(Rh, 3) + ns(PM_{2.5}, 3) + DOW + ns(time, 3/year) \quad (1)$$

where  $E(Y_t)$  is the expected number of deaths on day  $t$ ;  $HW$  represents a binary variable of heatwave;  $Rh$  is the daily mean relative humidity;  $PM_{2.5}$  refers to the concentration of  $PM_{2.5}$ ;  $ns$  is the natural cubic B spline function; the degrees of freedom ( $dfs$ ) are 3;  $DOW$  is a categorical variable representing the day of the week;  $ns(time, 3/year)$  refers to an  $ns$  function of time with 3  $dfs$  per summer to control for long-term trends (Fang et al., 2023). The selection of covariates and their parameters were consistent with that of previous studies (Fang et al., 2023; Hu et al., 2023a). In addition, we also further incorporated the daily maximum temperature into the model using an  $ns$  function with 3  $dfs$  to capture the added effect of heatwave (Yin et al., 2018).

In the second-stage analysis, a meta-analysis was conducted to calculate the pooled effects of heatwaves on mortality at the national level.

### 2.3.3. Estimating the modification of $O_3$ on heatwave-mortality associations

Analyses of both multiplicative and additive interactions were con-

ducted to comprehensively estimate the modification of  $O_3$  on heatwave-mortality associations. For multiplicative interaction analyses, the modification effect of  $O_3$  was examined by estimating the heatwave-mortality association during high and low  $O_3$  exposure. The models are as follows:

$$\text{Log}[E(Y_t)] = a + \beta_1(HW) + \beta_2(O_3) + \beta_3(HW:O_3) + ns(Rh, 3) + ns(PM_{2.5}, 3) + DOW + ns(time, 3/year) \quad (2)$$

In model (2),  $O_3$  is a binary variable, with 1 representing high  $O_3$  pollution ( $>50$ th  $O_3$  concentrations) and 0 representing low  $O_3$  pollution ( $\leq 50$ th  $O_3$  concentrations);  $HW:O_3$  is a multiplicative interaction item between  $HW$  and  $O_3$ .  $\beta_1$ ,  $\beta_2$ , and  $\beta_3$  are regression coefficients of  $HW$ ,  $O_3$ , and  $HW:O_3$ , respectively. The effects of heatwave on mortality during high  $O_3$  and low  $O_3$  exposures were evaluated to analyse the modifying effects of  $O_3$ , with both  $O_3$  and  $HW$  treated as dichotomous variables.

To further quantify the additive effects of co-exposure to heatwaves and high  $O_3$  pollution beyond the sum of their independent effects, a composite variable was included in the model. This variable includes four  $HW$ - $O_3$  events: non-heatwaves and low  $O_3$  pollution, heatwaves and low  $O_3$  pollution, non-heatwaves and high  $O_3$  pollution, and heatwaves and high  $O_3$  pollution. The non-heatwave and low  $O_3$  pollution event were used as the reference group.

$$\text{Log}[E(Y_t)] = a + \beta_4(HW - O_3) + ns(Rh, 3) + ns(PM_{2.5}, 3) + DOW + ns(time, 3/year) \quad (3)$$

where  $HW - O_3$  is a categorical variable, which represents heatwave- $O_3$  events;  $\beta_4$  is the effect of compound heatwave- $O_3$  event on mortality. The other parameters were identical to those in model (1).

The relative excess risk due to interaction (RERI), attributable proportion (AP), and synergy index (SI) were further used to assess the additive interaction (Correia and Williams, 2018; Knol et al., 2007). RERI measures the extent to which the combined effect of two exposure factors exceeds the sum of their individual effects when they coexist. AP represents the proportion of excess risk attributable to interaction in the group simultaneously exposed to both factors. SI quantifies the synergistic effect of the two exposure factors, reflecting the ratio of the combined effect to the sum of the independent effects. The calculation formulae for the three indicators are as follows:

$$RERI = RR_{11} - RR_{10} - RR_{01} + 1 \quad (4)$$

$$AP = \frac{RERI}{RR_{11}} \quad (5)$$

$$SI = \frac{RR_{11} - 1}{[(RR_{01} - 1) + (RR_{10} - 1)]} \quad (6)$$

where  $RR_{11}$  is the relative risk of co-exposure to heatwaves and high  $O_3$  pollution;  $RR_{10}$  is the relative risk of independent exposure to high  $O_3$  pollution, and  $RR_{01}$  is the relative risk of independent exposure to heat waves. If there is no additive interaction between heatwaves and  $O_3$ , the 95 % CI of RERI and AP should include 0, and the SI should include 1 (Knol et al., 2007). We used a Monte Carlo simulation to generate 1000 samples of the three interaction indices, and the 95 % confidence intervals were defined as the 2.5th and 97.5th percentiles of the simulated values.

#### 2.3.4. Mediation effect of $O_3$ on the heatwave-mortality association

We conducted a mediation analysis to estimate the direct and indirect effects of heatwaves on mortality (Alari et al., 2023). We decomposed the total effect (TE) of heatwaves on mortality into the direct effect (DE) and indirect effect (IE) mediated by  $O_3$ . To estimate the DE

and IE, we employed two regression models using the 'product method' established by Baron and Kenny (Baron and Kenny, 1986; Robins and Greenland, 1992; VanderWeele, 2016). Mediation effects were demonstrated when the following criteria were met: (1) the main independent variable (heatwave) was significantly related to the main dependent variable (mortality); (2) the independent variable (heatwave) was significantly associated with the mediator variable ( $O_3$ ); (3) the mediator variable ( $O_3$ ) was significantly related to the dependent variable (mortality) when the independent variable (heatwave) was controlled. In the mediation path, the potential confounders reported by previous studies, including RH,  $PM_{2.5}$ , day of week, and long-term trends, were all controlled during the three steps of mediation analysis to reduce the confounding bias (Bae et al., 2023; Gong et al., 2024).

The detailed process is shown in the Supplementary Materials, and a flowchart of the study is presented in Fig. S1.

#### 2.4. Sensitivity analysis

Based on previous studies, several similar sensitivity analyses were conducted to test the robustness of our results. First, we excluded  $PM_{2.5}$  in the model (Wen et al., 2022). We then changed the *dfs* (2–4 per year) for long-term trends (Fang et al., 2023). Third, we investigated whether our results were sensitive to other definitions of high  $O_3$  exposure and heatwaves in the model (Zhang et al., 2021a). Fourth, the two-pollutant models were constructed by respectively introducing  $PM_{10}$ , CO,  $NO_2$ , and  $SO_2$  into the model. The aim was to investigate whether the results remained stable after controlling for other pollutants (Chen et al., 2023; Li et al., 2021). Fifth, we incorporated the city level in the model to test the stability of the results. Sixth, we further conducted analyses of the impact of night-time heatwave (defined by daily minimum temperature  $\geq 92.5$ th lasting  $\geq 3$  consecutive days) and its co-occurrence with high  $O_3$  pollution. We also used ERA5-Land data matching with our death data as a sensitivity analysis.

All statistical analyses were performed using the R software (version 4.2.2, R Development Core Team). A two-tailed test with a *P*-value  $< 0.05$  was considered statistically significant.

### 3. Results

#### 3.1. Basic characteristics of the study sample

During the study period, 1,036,340 non-accidental deaths were recorded in the summer, with 58.38 % in males and 41.62 % in females. Of these, 438,184 (42.28 %), 191,673 (18.50 %), and 143,158 (13.81 %) deaths occurred because of CVD, CBD, and RESP, respectively (Table 1). The number of daily deaths during heatwave ( $9 \pm 6$ ) was higher than that during non-heatwave ( $8 \pm 6$ ). The values provided represent the means  $\pm$  standard deviations for daily mean air temperature ( $29.7 \pm 3.1$  °C) and daily 8-h maximum  $O_3$  ( $113.3 \pm 28.0$   $\mu\text{g}/\text{m}^3$ ) during heatwave, which were higher than those during non-heatwave (daily mean air temperature:  $24.7 \pm 3.7$  °C; daily 8-h maximum  $O_3$ :  $87.3 \pm 26.3$   $\mu\text{g}/\text{m}^3$ ) (Table S2). More detailed information on the study sample is shown in Fig. S3–S5 and Table S2. The correlation coefficients between the meteorological variables and air pollutant concentrations are shown in Fig. S6. Positive associations were noted between temperature and air pollutant concentrations.

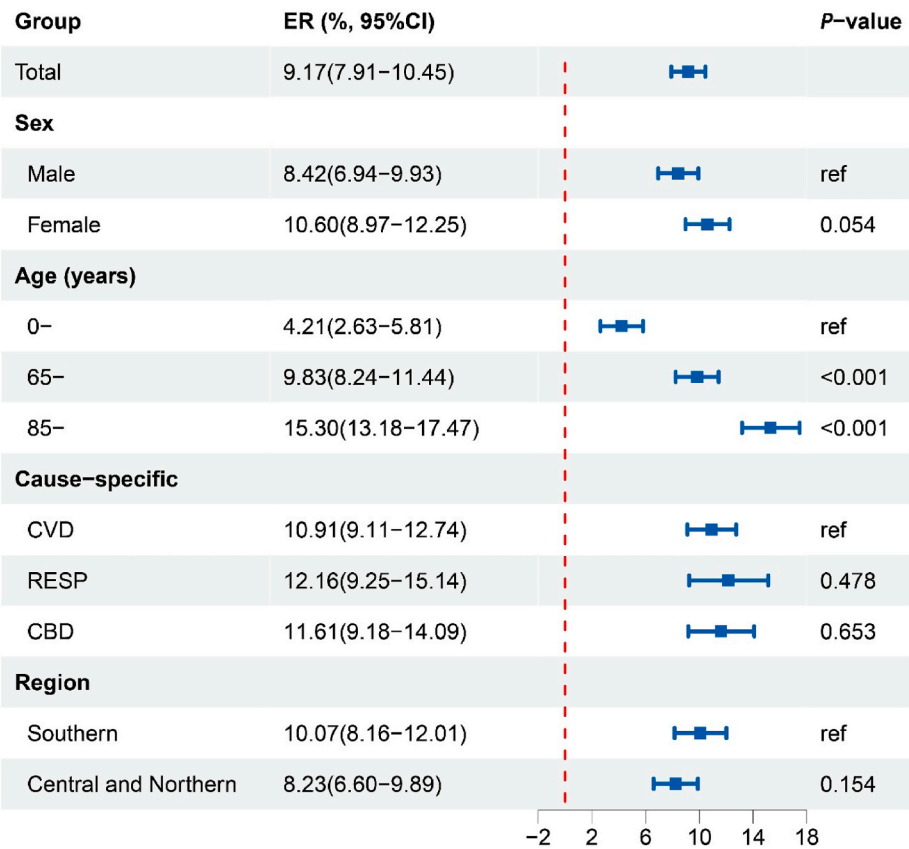
#### 3.2. The association of heatwave with mortality

As shown in Fig. 2, we found that heatwaves increased mortality risk (ER = 9.17 %, 95 % CI: 7.91–10.45 %), with a marginally significant increase in risk for females (ER = 10.60 %, 95 % CI: 8.97–12.25 %) than for males (ER = 8.42 %, 95 % CI: 6.94–9.93 %), and for individuals aged  $\geq 85$  years (ER = 15.30 %, 95 % CI: 13.18–17.47 %) compared to



**Table 1**  
Characteristics of mortality and environmental factors in 278 study counties/districts in China.

| Group                                  | Number of deaths | Proportion (%) | Mean (Sd)     | Min   | P <sub>25</sub> | P <sub>50</sub> | P <sub>75</sub> | Max    |
|----------------------------------------|------------------|----------------|---------------|-------|-----------------|-----------------|-----------------|--------|
| <b>Total</b>                           | 1036340          | 100.00         | 8 (6)         | 0     | 4               | 7               | 11              | 129    |
| <b>Sex</b>                             |                  |                |               |       |                 |                 |                 |        |
| Males                                  | 605019           | 58.38          | 5 (4)         | 0     | 2               | 4               | 6               | 73     |
| Females                                | 431269           | 41.62          | 3 (3)         | 0     | 1               | 3               | 5               | 56     |
| <b>Age (years)</b>                     |                  |                |               |       |                 |                 |                 |        |
| 0–64                                   | 271228           | 26.17          | 2 (2)         | 0     | 1               | 2               | 3               | 40     |
| 65–84                                  | 541408           | 52.24          | 4 (3)         | 0     | 2               | 4               | 6               | 73     |
| ≥85                                    | 223703           | 21.59          | 2 (2)         | 0     | 0               | 1               | 3               | 46     |
| <b>Cause-specific</b>                  |                  |                |               |       |                 |                 |                 |        |
| CVD                                    | 438184           | 42.28          | 3 (3)         | 0     | 1               | 3               | 5               | 69     |
| RESP                                   | 143158           | 13.81          | 1 (1)         | 0     | 0               | 1               | 2               | 30     |
| CBD                                    | 191673           | 18.50          | 2 (2)         | 0     | 0               | 1               | 2               | 34     |
| <b>Region</b>                          |                  |                |               |       |                 |                 |                 |        |
| Southern                               | 617053           | 60.00          | 8 (6)         | 0     | 4               | 7               | 11              | 129    |
| Central and Southern                   | 419287           | 40.00          | 8 (5)         | 0     | 4               | 7               | 10              | 46     |
| <b>Meteorological factors</b>          |                  |                |               |       |                 |                 |                 |        |
| Daily mean air temperature (°C)        | –                | –              | 25.65 (4.05)  | 8.70  | 22.53           | 26.19           | 28.72           | 37.97  |
| Daily relative humidity (%)            | –                | –              | 78.62 (10.07) | 28.70 | 73.50           | 80.09           | 85.67           | 99.37  |
| <b>Air pollutants</b>                  |                  |                |               |       |                 |                 |                 |        |
| O <sub>3</sub> (µg/m <sup>3</sup> )    | –                | –              | 92.14 (28.50) | 12.39 | 71.49           | 88.12           | 107.75          | 312.18 |
| PM <sub>2.5</sub> (µg/m <sup>3</sup> ) | –                | –              | 26.64 (12.91) | 4.55  | 17.71           | 23.24           | 31.90           | 215.75 |
| CO (mg/m <sup>3</sup> )                | –                | –              | 0.84 (0.27)   | 0.25  | 0.67            | 0.79            | 0.94            | 8.52   |
| NO <sub>2</sub> (µg/m <sup>3</sup> )   | –                | –              | 18.60 (7.20)  | 4.06  | 13.61           | 16.47           | 21.61           | 87.54  |
| SO <sub>2</sub> (µg/m <sup>3</sup> )   | –                | –              | 14.52 (5.90)  | 2.28  | 9.82            | 13.67           | 18.42           | 62.42  |
| PM <sub>10</sub> (µg/m <sup>3</sup> )  | –                | –              | 45.57 (18.52) | 9.66  | 32.77           | 40.86           | 53.33           | 275.18 |

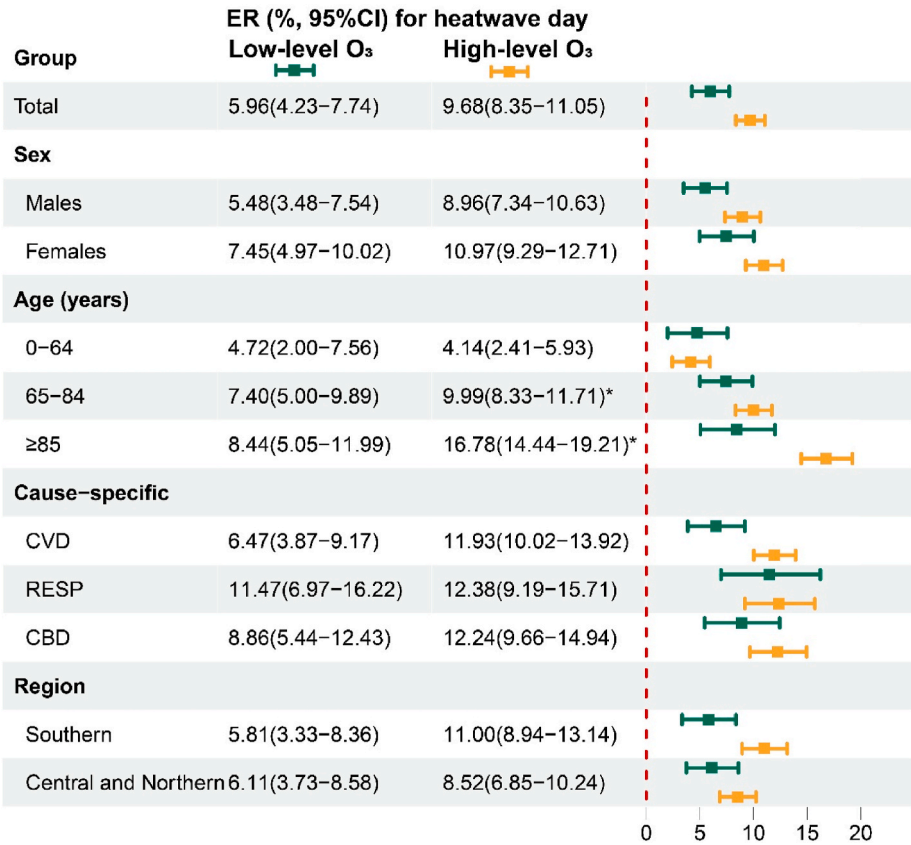


**Fig. 2.** Excess risk (ER, %) of mortality attributed to heatwave by age, sex, cause of death and region.

younger populations. For different causes of death and different regions, the ERs for heatwave were similar. We further stratified heatwave-related mortality risk by O<sub>3</sub> quartile and found a clear, monotonic increase in mortality risk with rising O<sub>3</sub> concentrations (Table S8). The added effect of the heatwave accounted for approximately one-quarter of the total effect, with increased the mortality risk by 2.32 % (95 % CI: 1.15–3.50 %) (Table S3).

3.3. Modification effects of O<sub>3</sub> on heatwave-mortality associations

As shown in Fig. 3, O<sub>3</sub> modified the association between heatwaves and mortality. For total population, heatwave had a higher mortality risk during high O<sub>3</sub> pollution (ER = 9.68 %, 95 % CI: 8.35–11.05 %) compared with low O<sub>3</sub> pollution (ER = 5.96 %, 95 % CI: 4.23–7.74 %). The stratified analysis indicated that heatwave related mortality risks



**Fig. 3.** Comparison of excess risks (ER, %) of mortality attributed to heatwave during high-level O<sub>3</sub> and low O<sub>3</sub> pollution by sex, age, cause of death and region. **Note:** \* indicated that the difference between subgroups was statistically significant, with males, people aged 0–64 years old, CVD patients, residents in Southern regions as reference.

increased as aging, while we did not observe significant differences between sex, cause-specific, and regions.

Fig. 4 and Table S4 show the additive effects of O<sub>3</sub> and heatwave on mortality. We found that the RERI was 0.03 (95 % CI: 0.01–0.06), and the SI was 1.33 (95 % CI: 1.02–1.83), indicating the presence of additive interactions between O<sub>3</sub> and heatwaves. The AP was 2.70 % (95 % CI: 0.20–5.20 %), indicating that the additive effect of heatwaves and high O<sub>3</sub> pollution co-exposure accounted for 2.70 % of the TE. After stratified analysis, we observed that females and the elderly aged 65 and above had higher mortality risk when concurrently exposure to heatwave and high O<sub>3</sub> pollution, but we could not observe significant differences between various cause and regions. We further found that the effect sizes increased as the thresholds (daily maximum air temperature and duration of exceeding the temperature threshold) of heatwave rose. For instance, when the heatwave threshold was set at the 90th, 92.5th, and 95th percentiles for 2 days, the interaction effects observed were 1.30, 2.10 and 3.40 %, respectively. When we defined a heatwave as 2, 3, and 4 consecutive days exceeding the 95th percentile of maximum air temperature, the AP values were 3.40, 4.40 and 5.30 %, respectively.

3.4. Mediation effect of O<sub>3</sub> on the heatwave-mortality association

For total population, O<sub>3</sub> significantly mediated the association between heatwave and mortality, with 9.33 % (95 % CI: 7.17–11.76 %) of effect being mediated. When stratified by sex, the mediation effect was greater for females (9.91 %) than for males (8.71 %). In terms of age groups, individuals aged ≥85 years exhibited a higher mediation effect (11.48 %) compared to other age groups. CVD, RESP, and CBD had similar mediating effects on the causes of death (Fig. 5). We observed a higher but non-significant mediated proportion in the central and northern region than southern regions. More detailed information is

provided in Fig. S7.

3.5. Sensitivity analysis

A series of sensitivity analyses were conducted, including variations without adjustment for PM<sub>2.5</sub> and other pollutants (PM<sub>10</sub>, CO, NO<sub>2</sub>, SO<sub>2</sub>), changes to the *dfs* (2–4 per year) for long-term trends, adjustments to the definition of high O<sub>3</sub> pollution, and modifications to the heatwave definition. Additionally, the city level was incorporated in the model. All results were consistent with the main findings, suggesting that our results were robust. Moreover, we used the daily minimum temperature to define nighttime heatwaves and further estimate their effect. The results indicated that the ER of nighttime heatwave on mortality was 6.00 % (95 % CI: 4.91–7.11 %), which was slightly lower than that of daytime heatwave (ER = 9.17 %, 95 % CI: 7.91–10.45 %) (Table S7). Consistent with the daytime heatwave, the nighttime heatwave had a greater mortality risk during high O<sub>3</sub> pollution than during low O<sub>3</sub> pollution (Table S5 and S6). In order to test the impact of source of meteorological data, we conducted sensitive analysis used ERA5 and ERA5-Land data separately and observed consistent results (Table S5).

4. Discussion

Climate change increases the probability of encountering heatwave-O<sub>3</sub> events, potentially escalating the associated health risks. This comprehensive study in China examined the modifying and mediating influence of O<sub>3</sub> on the relationship between heatwaves and mortality. Our results indicate that elevated O<sub>3</sub> concentrations amplified the impact of heatwaves on mortality, with O<sub>3</sub> accounting for 9.33 % of this effect. These insights could enhance our understanding of the factors contributing to increased mortality during heatwaves and underscore

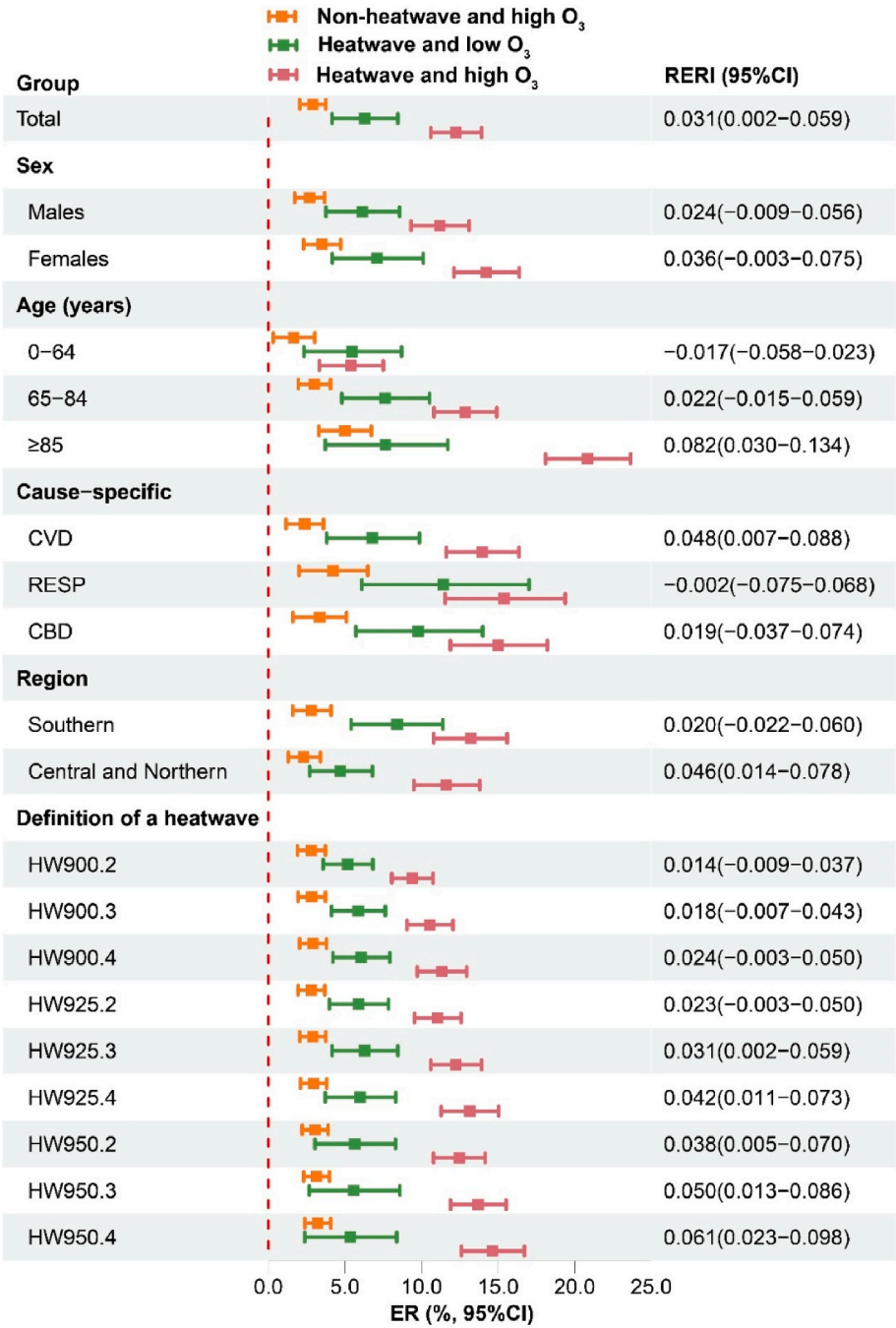


Fig. 4. Additive interactions of O<sub>3</sub> and heatwave on mortality for sex, age, cause of death, region and different thresholds for heatwave.

the need for coordinated management of climate warming and O<sub>3</sub> pollution in the context of climate change.

Our study found that the ER of heatwaves on total mortality during periods of high O<sub>3</sub> was almost twice as high as that during periods of low O<sub>3</sub> (ER: 9.68 % vs. 5.96 %). This finding is consistent with those of several previous studies (Du et al., 2024; Qi et al., 2023; Xu et al., 2023). For instance, a national study based on 250 counties in China found a higher risk of all-cause mortality from co-exposure to heatwaves and high O<sub>3</sub> pollution events than from independent exposure to heatwaves (RR: 1.29 vs. 1.22) (Du et al., 2024). Another meta-analysis also observed that heatwave-related ERs for non-accidental mortality were 9.30 % under low O<sub>3</sub> pollution and 11.40 % under high O<sub>3</sub> pollution (Hu et al., 2022). Moreover, we observed that a 2.70 % additive mortality risk could be attributed to the heatwave-O<sub>3</sub> event, rather than the sum of

their independent effects, in the total population. This interaction was much higher for individuals aged ≥85 years (6.80 %) and patients with CVD (4.20 %), although it was lower than the findings of a previous study in Jiangsu province of China (6.00–36.00 %) (Xu et al., 2023). The differences among the various studies may be explained by variations in statistical methods, study period, study area, participants, exposure assessment, and heatwave definition. First, the air conditioning penetration rate and reduction in outdoor activities may partially explain this difference. O<sub>3</sub> is a reactive chemical found at much lower concentrations indoors than outdoors (Gryparis et al., 2004). Residents with air conditioners tend to stay indoors to avoid high temperatures, resulting in lower exposure to O<sub>3</sub> and heat. Additionally, the varying adaptation capacities of populations to O<sub>3</sub> and heat in different regions may also be a contributing factor. Although some studies on physiological

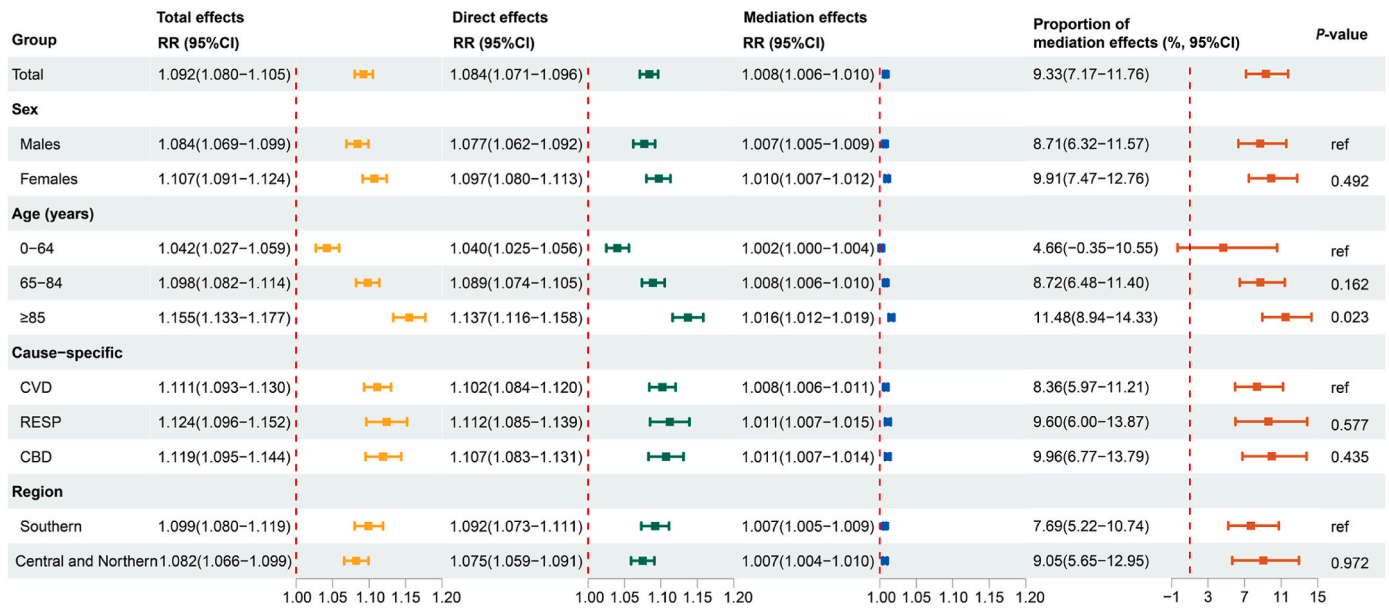


Fig. 5. Mediation effect of  $O_3$  on the association between heatwave and mortality.

adaptation (Folinsbee et al., 1994; Ren et al., 2009) have suggested that populations living in areas with high  $O_3$  concentrations and heat may be less sensitive to day-to-day variations in  $O_3$  concentrations and temperature, other studies have found that populations living in highly exposed areas may be more sensitive (Huang et al., 2024). However, in both cases, this could lead to inconsistencies in the risk across different studies. Third, several studies have utilised case-crossover designs to explore the relationship between  $O_3$ , heat waves, and mortality (Xu et al., 2023), which differs from our ecological time-series study approach.

The synergistic effects of heatwaves and  $O_3$  can be explained through potential biological mechanisms. Evidence from previous studies indicates that  $O_3$  may directly affect the respiratory tract through inhalation and cause systemic inflammation and oxidative stress (Arjomandi et al., 2015), which affect autonomic nervous system regulation, thus increasing people's susceptibility to high temperatures. Therefore, an increase in  $O_3$  concentration during heatwave periods and the aforementioned physiological mechanisms may jointly contribute to synergistic health effects.

We also found that the combined mortality risk from heatwaves and high  $O_3$  pollution increased with the intensity and duration of the heatwaves, which is similar to the results of a previous study (Xu et al., 2023). Although the underlying mechanisms are not yet fully understood, these results may be related to several physiological processes. First, when individuals are exposed to heatwaves or high ambient temperatures, the core body temperature rises, and the thermoregulatory system accelerates the transfer of heat to the environment (Yousef et al., 2024). Longer duration and higher severity of heatwaves could lead to a greater increase in the respiratory rate (Berihulay et al., 2019), resulting in greater inhalation of  $O_3$  and consequently an increased mortality risk. Secondly,  $O_3$  is a strong oxidant that can undergo oxidative reactions with biological tissue components (Miller et al., 2016). With the extension of heatwave exposure, these oxidative reaction processes may be expedited, further augmenting health damage. Finally, as the intensity and duration of heatwaves increase, the body may surpass its thermoregulatory ability, triggering systemic inflammation, oxidative stress, and endothelial dysfunction (Alahmad et al., 2020), making it more arduous for the body to defend itself against  $O_3$ -induced damage.

We further identified that females and older individuals were more vulnerable to events involving both heatwaves and high  $O_3$  pollution,

which is consistent with previous studies (Du et al., 2024; Qi et al., 2023; Xu et al., 2023). Gender inequality in education, employment, and access to resources may contribute to a higher mortality risk for females (Roy et al., 2022). Differences in hormones and structural/morphological characteristics of the respiratory system may also contribute to the different risks between male and female individuals (Bell et al., 2014; Harms, 2006). In addition, heat stress is more likely to trigger frequent circulatory and thermoregulatory issues in females (Burse, 1979), while  $O_3$  exposure can lead to a stronger respiratory inflammatory response. Older individuals are more vulnerable to heatwave- $O_3$  events owing to reduced temperature regulation capacity, decreased chest wall compliance, weakened respiratory muscle strength, lower vital capacity, and medication interference with normal physiological processes (Basagaña et al., 2011; Janssens, 2005; Schifano et al., 2009). Moreover, the decline in physical and cognitive function affects older individuals' ability to access adaptive technologies, such as moving to cooler areas, taking cool baths, and wearing suitable clothing (Xi et al., 2024). Given that air conditioning is an effective heat adaptation strategy (Sera et al., 2020), the utilisation of air conditioners may significantly influence age and sex disparities. A prior study conducted in Hong Kong (China) revealed that females tend to use air conditioning less often, with 56.55 % of females keeping the air conditioner 'on', compared to 65.48 % of males (Gao et al., 2024). Additionally, the study demonstrated a positive correlation between age and the frequency of keeping the air conditioner 'off'.

In previous studies on the heatwave-mortality association, researchers usually conceptualised the role of  $O_3$  as a confounder or effect modifier (Du et al., 2024; Reid et al., 2012; Xu et al., 2023). Several recent studies have demonstrated that  $O_3$  plays an important role in the causal pathway between heatwaves and mortality (Alari et al., 2023; Bae et al., 2023). For instance, a French study found that  $O_3$  accounted for 2.00–31.00 % of heatwave mortality in 15 urban cities; however, epidemiological evidence in China is still insufficient. In this study, we observed a 9.33 % mediating effect of  $O_3$  on heatwaves and total mortality. The differential mediation effects across studies may be attributed to variations in  $O_3$  concentrations and study populations (Hoesly et al., 2018; Hu et al., 2023b). Moreover,  $O_3$  played a more significant role in the heatwave-mortality association among individuals aged  $\geq 85$  years, potentially owing to their susceptibility to  $O_3$ . Furthermore, we observed similar mediating effects of  $O_3$  among individuals with CVD, CBD and RESP. Functionally, the cardiovascular and pulmonary systems



interact closely and jointly maintain blood circulation and gas exchange (Dong et al., 2014). They also share the similar biological mechanisms related to heatwaves and O<sub>3</sub>, such as systemic inflammation and oxidative stress (Cheng et al., 2024; Goodman et al., 2015; Wiegman et al., 2020). Our findings suggest that reducing outdoor activities during heatwaves to minimize exposure to O<sub>3</sub> could be an effective strategy for mitigating heatwave-related effects in older individuals. Our findings contribute to a better understanding of the mechanisms underlying the observed heatwave-mortality effects.

Our findings provide valuable information for the coordination of climate change and air pollution management strategies. From a theoretical perspective, these findings deepen our understanding of the mechanism of the association between heatwaves and mortality, namely, the modification and mediation of O<sub>3</sub> on the heatwave-mortality association. From an application perspective, owing to the mediation of O<sub>3</sub> on the mortality effects of heatwaves, if the pathways between heatwaves and O<sub>3</sub> are partially blocked by reducing O<sub>3</sub> precursors, the mortality risk of heatwaves could be considerably reduced. In addition, our results indicate that it is necessary to develop an early warning system for heatwave-O<sub>3</sub> events for risk communication and behavioural adaptation, such as avoiding outdoor activities during heatwaves, which will reduce the mortality risk of heatwave-O<sub>3</sub> event.

This study had several strengths. First, this was a large-scale study that included data on 1,036,340 deaths in five provinces in China. Second, our findings provide novel and quantitative evidence of the modifying and mediating effects of O<sub>3</sub> on heatwave-mortality associations. This study also had certain limitations. First, this was an ecological study, which is subject to ecological fallacy. Therefore, the findings should be interpreted with caution. Second, owing to data restrictions, we were unable to include health data from Northwest China, and the study locations were not evenly distributed within the regions, which may limit the generalizability of our findings. Moreover, confining the study period to 2013–2017 might not fully elucidate the role of O<sub>3</sub>, as the overall O<sub>3</sub> concentration in China has shown a significant increase since 2018 (Wang et al., 2022). Third, we employed data on heatwaves and O<sub>3</sub> concentrations at the county/district level as the population exposure and constructed a time-series design, rather than an individual-level exposure design such as a case-crossover study, where each subject serves as their own control within the same month. Therefore, some potential bias in individual level could not be avoided. Fourth, the Baron and Kenny method was used, which does not allow for incorporation of interactions between heatwave and O<sub>3</sub> on mortality. Although causal mediation analysis would have been preferable, technical constraints associated with the DLNM with quasi-Poisson link function, as well as the time-series characteristics of the data, restricted its implementation. Therefore, this traditional mediation analysis was commonly used in previous ecological studies (Alari et al., 2023; Bae et al., 2023; Gong et al., 2024). Finally, the current study did not estimate the lag effect of heatwaves and O<sub>3</sub> on mortality using the distributed lag non-linear model (DLNM), as DLNM could not account for the lag effect of the mediator in our mediation analysis (Bae et al., 2023).

## 5. Conclusions

The nationwide study presented novel insights into the increased mortality risk associated with the coexistence of heatwaves and high O<sub>3</sub> pollution. High O<sub>3</sub> pollution exacerbates the effect of heatwave on mortality, and a 2.70 % additive mortality interaction between O<sub>3</sub> and heatwave was observed. Females and older individuals are more vulnerable toward this compound exposure. Moreover, O<sub>3</sub> was found to contribute to 9.33 % of the effect of heatwaves on mortality, and this mediation proportion was higher among individuals aged 65 years and above. Our results provide evidence that O<sub>3</sub> may function in a dual capacity: partly mediating the effect of heatwave on mortality and partly modifying the strength of this association. This dual role reflects the complex interplay between meteorological conditions, air quality, and

health outcomes. These findings highlighted the need to coordinate the management of global warming and air pollution, as well as to implement adaptation strategies to alleviate their health impact.

## CRediT authorship contribution statement

**Jianxiong Hu:** Writing – original draft, Software, Methodology, Formal analysis. **Wei quan Zeng:** Writing – original draft, Visualization, Methodology, Formal analysis. **Yanfang Guo:** Writing – original draft, Methodology, Conceptualization. **Ruilin Meng:** Visualization, Validation, Methodology. **Shaoli Huang:** Visualization, Validation, Software. **Chunliang Zhou:** Validation, Resources, Methodology, Investigation, Data curation. **Yize Xiao:** Visualization, Validation, Resources, Methodology, Investigation. **Min Yu:** Visualization, Validation, Resources, Methodology, Data curation. **Biao Huang:** Visualization, Validation, Resources, Data curation. **Dalin Lu:** Writing – review & editing, Methodology, Conceptualization. **Guanhao He:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Wenjun Ma:** Writing – review & editing, Project administration, Methodology, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envpol.2025.126493>.

## Data availability

Data will be made available on request.

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